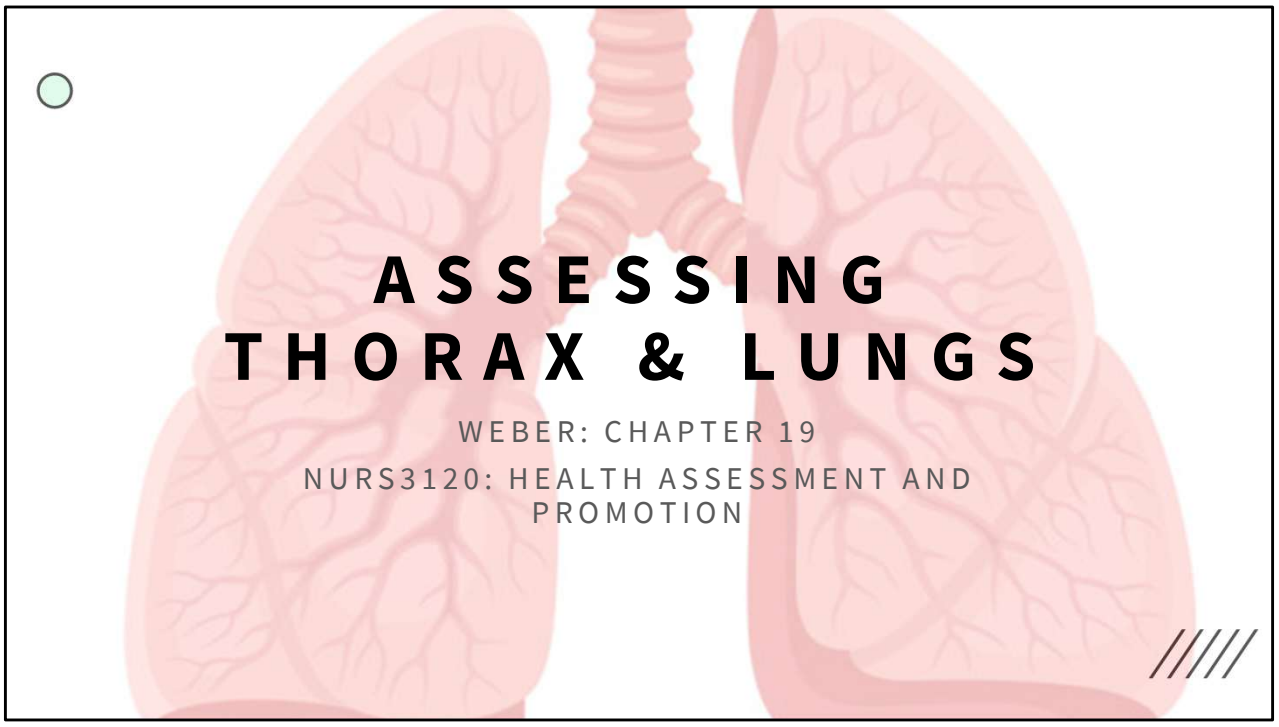


ASSESSING THORAX & LUNGS

WEBER: CHAPTER 19

NURS3120: HEALTH ASSESSMENT AND PROMOTION

Objectives

- Describe the function and structure of the thorax and lungs
- Identify the thoracic landmarks in relation to the underlying thoracic structures
- Discuss the risk factors for lung cancer and ways to reduce one's risks
- Differentiate normal vs. abnormal findings of the thorax and lungs
- Analyze the client's interview and physical assessment data related to the thorax and lungs to formulate valid clinical judgments



Function of the respiratory system

01

Supplying oxygen to the body for energy production

02

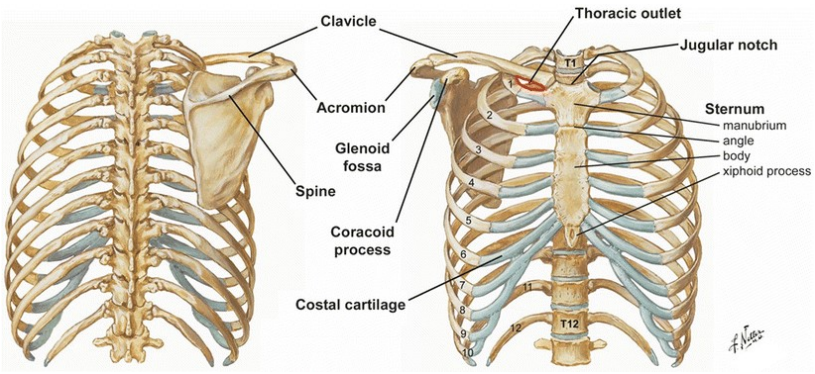
Removing carbon dioxide (CO_2) as a waste product of energy reactions

03

Maintaining homeostasis (acid-base balance) of arterial blood

The purpose of respiration is to maintain an adequate oxygen level in the blood to support cellular life. By providing oxygen and eliminating carbon dioxide, respiration assists in the rapid compensation for metabolic acid–base defects. However, changes in the respiratory pattern can cause acid–base imbalances.

Anatomy of the thorax

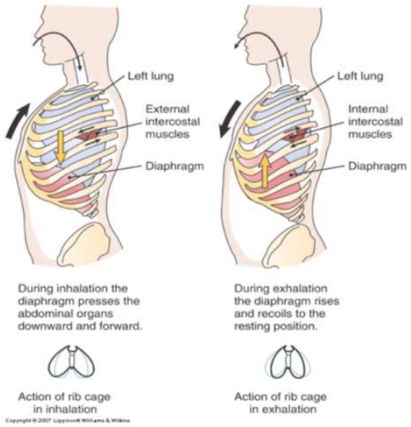


The thoracic cage is constructed of the sternum, 12 pairs of ribs, 12 thoracic vertebrae, muscles, and cartilage. It provides support and protection for many important organs, including those of the lower respiratory system. The lungs are two cone-shaped, elastic structures suspended within the thoracic cavity. The apex of each lung extends slightly above the clavicle. The base is at the level of the diaphragm.

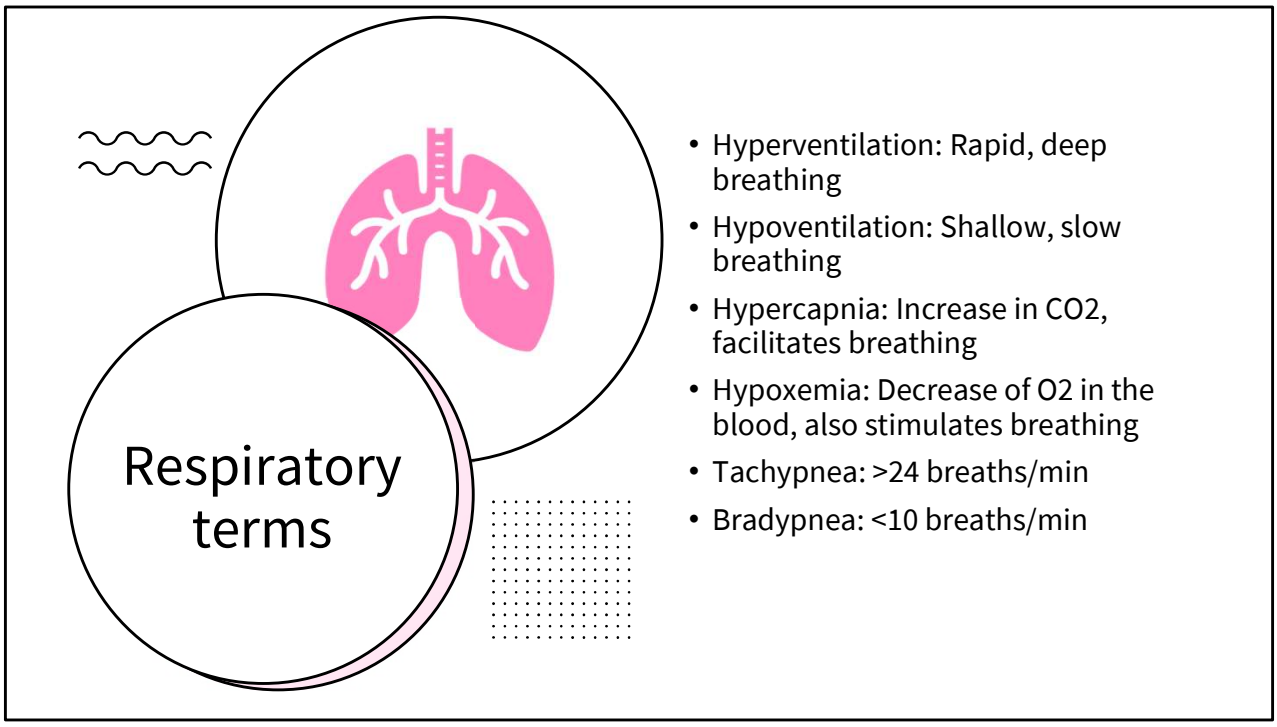
The thoracic cavity is lined by a thin, double-layered serous membrane referred to as the pleura. The parietal pleura lines the chest cavity, and the visceral pleura covers the external surfaces of the lungs. The pleural space lies between the two pleural layers. In the healthy adult, the lubricating serous fluid between the layers allows movement of the visceral layer over the parietal layer during ventilation without friction. Because the pleural space is one of the physiologic third spaces for body fluid storage, severe dehydration will reduce the volume of pleural fluid, resulting in the increased transmission of lung sounds and a possible friction rub.

Mechanism of breathing

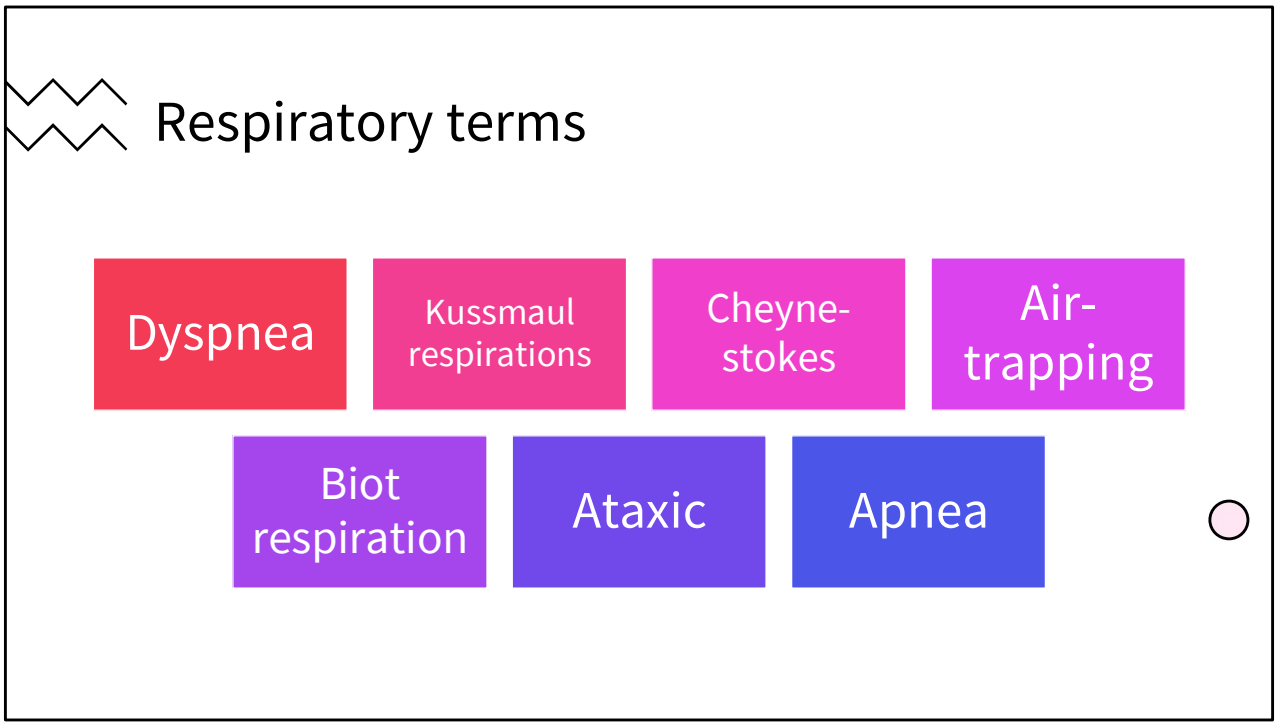
- Purpose
  - To maintain an adequate oxygen level in the blood to support cellular life
- Breathing patterns change according to cellular demands
- Involuntary control



Breathing patterns change according to cellular demands often without awareness on the part of the individual. Such involuntary control of respiration is the work of the medulla and pons located in the brainstem. The hypothalamus and the sympathetic nervous system also play a role in involuntary control of respiration in response to emotional changes, such as fear or excitement. Hormonal regulation, changes in oxygen or carbon dioxide levels in the blood, or changes in the hydrogen ion pH level cause changes in breathing patterns.



Normal respiratory rate is: 12-20 breaths/min and regular
Tachypnea is more than 24 breaths/min (Fever, anxiety, exercise)
Bradypnea is less than 10 breaths/min (potentially normal in well-conditioned athletes, medication-induced, neurologic coma)
Hyperventilation is increased rate and increased depth. Usually occurs with extreme exercise, fear, or anxiety. Exhaling more carbon dioxide then inhaling oxygen.
Hypoventilation. Carbon dioxide is not being exchanged properly and remains in the blood stream. Usually due to overdose of narcotics or anesthetics.
Hypercapnia: Elevated carbon dioxide levels in the blood (hypoventilation, COPD, sleep apnea)
Hypoxemia: low levels of oxygen in the blood…



Dyspnea- shortness of breath

Kussmaul respirations- Rapid, deep, labored breathing. Described as air hunger. Body is trying to expel large amount of CO2- Associated with diabetic ketoacidosis

Cheyne-Stokes- abnormal breathing pattern with cycles of gradually increasing (crescendo) and then decreasing (decrescendo) breaths, followed by a temporary stop in breathing (apnea), repeating every 30 seconds to 2 minutes, often seen in severe heart failure, stroke, or at the end of life, caused by delays in the brain's CO2 detection.

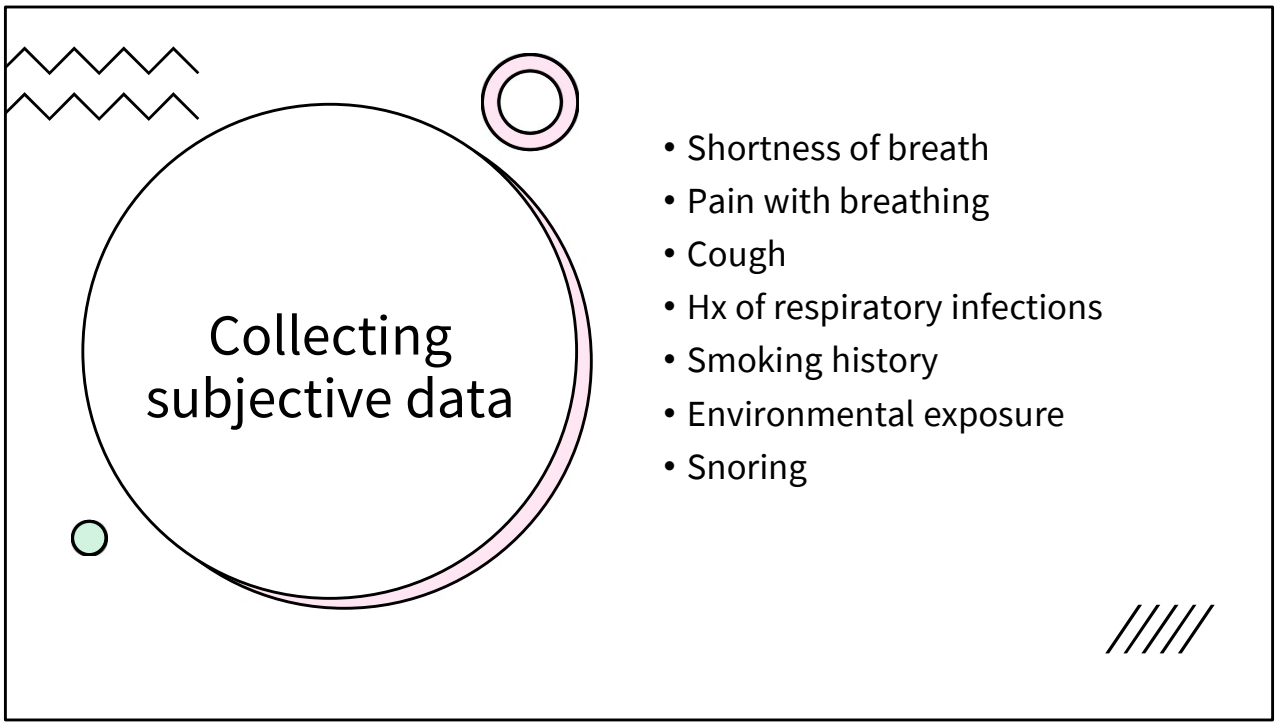
Air-trapping- retention of air, not able to completely exhale. COPD.

Biot respirations: Irregular pattern characterized by varying depth and rate of respirations followed by periods of apnea. (Meningitis or severe brain damage)

Ataxic: Significant disorganization with irregular and varying depths of respiration. (A more extreme expression of Biot respirations indicating

respiratory compromise)

Apnea- suspension of breathing



Dyspnea: Could be pulmonary disorders like COPD, asthma, pneumonia, CHF, etc) COLDSPA

Chest pain: Need to determine of cardiac, pulmonary, GI, musculoskeletal

Cough: Continuous usually acute infection. Early morning typically chronic bronchial inflammation or smoking. Evening/night can be exposure to irritants during the day or postnasal drip. Should ask: Productive? What does sputum look like? Any hemoptysis (bloody sputum)? This is seen with more serious respiratory conditions.

History of infections or lung issues? Useful to know due to possible scarring of lung tissue or potential changes in the appearance of the thorax or respiratory sounds.

Smoking: #1 risk factor for lung cancer. Also vaping/e-cigarettes. How much? How long? Ever tried to quit? Do they currently want to quit?

Environmental exposure: Environmental agents include coal dust, insecticides,

paint, pollution, asbestos fibers, and the like. Exposure could result in increased incidence of respiratory conditions.

Do you snore while you sleep??? Sleep apnea

Older adult health history questions

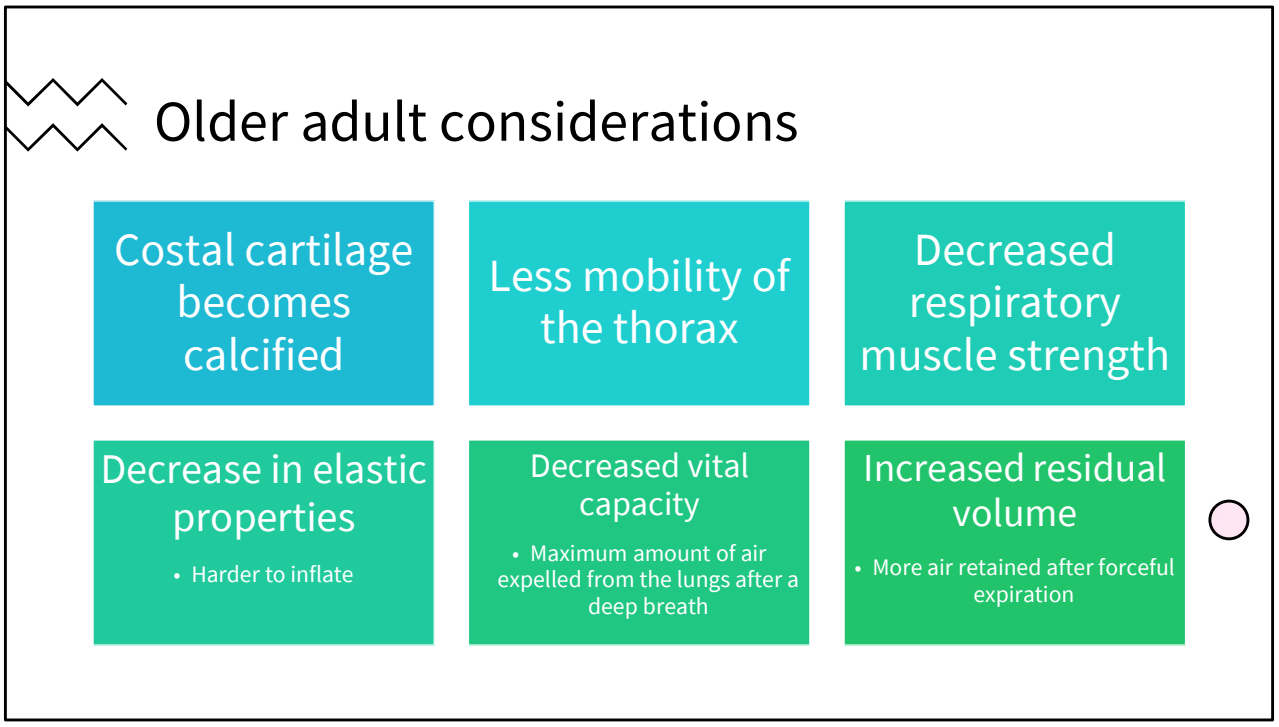
- Any shortness of breath or fatigue with daily activity? (DUE: Dyspnea upon exertion)
- What is your usual amount of physical activity?
- Any chest pain or cough with breathing?
- How many pillows do you sleep on, or do you sleep in a recliner/chair? (Orthopnea)
- Do you awaken from sleep with SOB? (PND: Paroxysmal nocturnal dyspnea)



Dyspnea- shortness of breath
Older adults may experience dyspnea with certain activities due to age-related changes of the lungs (loss of elasticity, fewer functional capillaries, and loss of lung resiliency).
Dyspnea will occur with typical nonstrenuous activities such as walking one block or climbing two stairs in clients with lung disease or congestive heart failure.
Chest pain is a medical emergency to rule out cardiac ischemia, but if there are other symptoms with the chest pain like a cold, fever, or deep breathing. This is why COLDSPA is important.

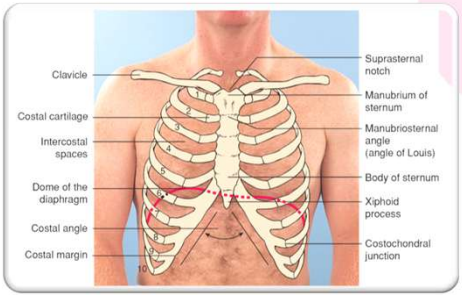
Orthopnea- difficulty breathing when supine. Seen with COPD, heart failure.

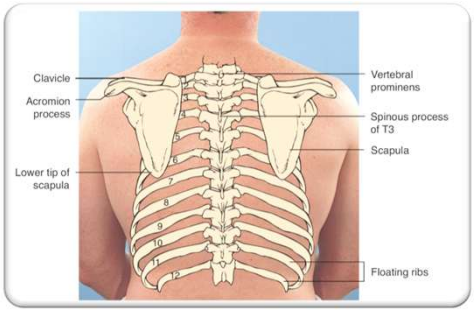
PND: sudden, severe shortness of breath and coughing that wakes you from sleep, typically 1-2 hours after falling asleep, often feeling like suffocation and relieved only by sitting or standing upright. Seen with left-sided heart failure. When lying down, heart cannot pump effectively, fluid builds up in lungs.



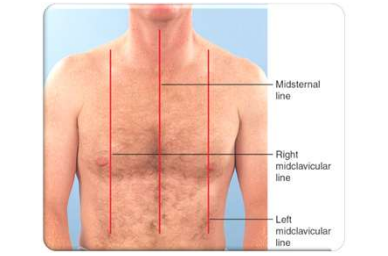
The ability to cough effectively may be decreased in the older client because of weaker muscles and increased rigidity of the thoracic wall.
Because of calcification of the costal cartilages and loss of the accessory musculature, the older client's thoracic expansion may be decreased, although it should still be symmetric.
Decreased mobility of the lungs is a normal finding in the older adults

ANATOMICAL LANDMARKS





ANATOMICAL LANDMARKS





Collecting objective data

- Inspection
  - Nasal flaring
  - Pursed-lip breathing
  - Color/shape of nails
  - Color of face, lips, chest

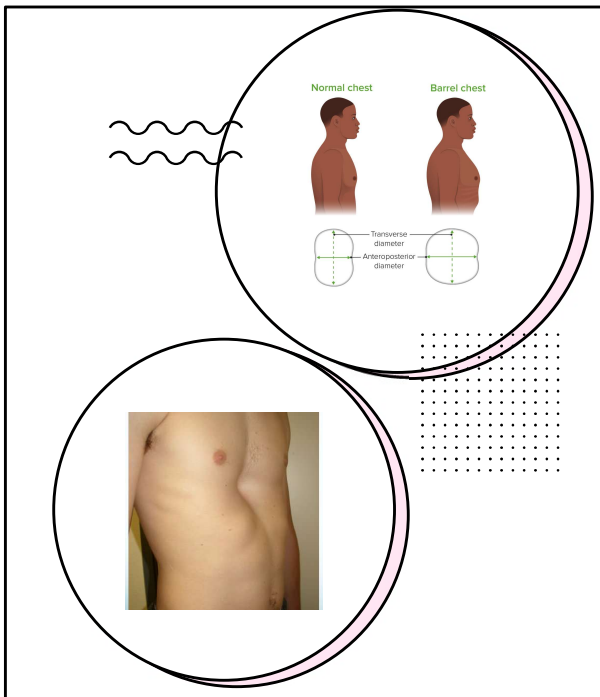




Nasal flaring is abnormal and seen with labored respirations (especially in small children) and can indicate hypoxia; Pursed lip breathing may be seen in asthma, emphysema, or CHF as a physiologic response to help slow down expiration and keep alveoli open longer.

Nails: Normally have a pink tone and a 160-degree angle between the nail base and the skin. Abnormal clubbing is a 180-degree or greater angle. Abnormal color is pale or cyanotic nails that may indicate hypoxia.
When observing the face, lips and chest of a client: Normal is evenly colored skin tone, without unusual or prominent discoloration. Abnormal is ruddy to purple complexion may seen with COPD or CHF as a result of polycythemia: Polycythemia is a disease state in which the hematocrit is elevated.. This develops as a response to chronic hypoxemia, which triggers increased production of erythropoietin by the kidneys. Cyanosis (Blue/dull) color change may be seen if the client is hypoxic or cold.

HTTPS://www.lung.org/lung-health-and-diseases/lung-disease-lookup/copd/patient-resources-and-videos/pursed-lip-breathing-video.html



Anterior assessment of the thorax

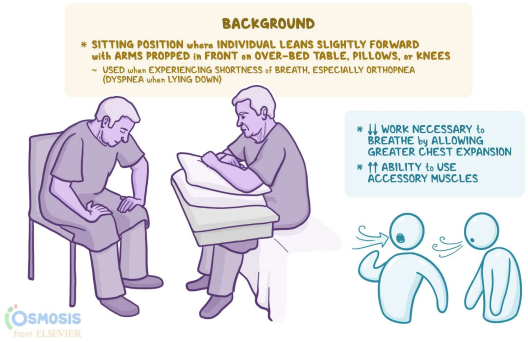
- Inspect
  - Shape & configuration
  - Position of the sternum
  - Slope of the ribs
  - Quality & pattern of respiration
  - Intercostal spaces (between ribs)
  - Use of accessory muscles

When you inspect for shape and configuration; have the client sit with arms at the sides. Stand in front of the client. The anteroposterior diameter is less than the transverse diameter usually 1:2. Abnormal is equal diameter, resulting in a barrel chest. Clients with emphysema often have a barrel chest because of hyperinflation of the lungs.
Position of the sternum should be midline and straight.
Abnormal: pectus excavatum: sunken sternum or funnel chest
Slope of the ribs: Downward with symmetric intercostal spaces. Abnormal: barrel chest….. The costal angle of more than 90 degrees
Quality and pattern of respirations: We all know this right???? Rate, rhythm, and depth
Intercostal spaces: Ask the patient to breathe normally and observe the intercostal spaces: normal: no retractions or bulging: Abnormal: retraction of the intercostal space indicates an increased inspiratory effort. Bulging of intercostal spaces indicates trapped air such as emphysema or asthma
Use of accessory muscles: normal is no use of accessory muscles

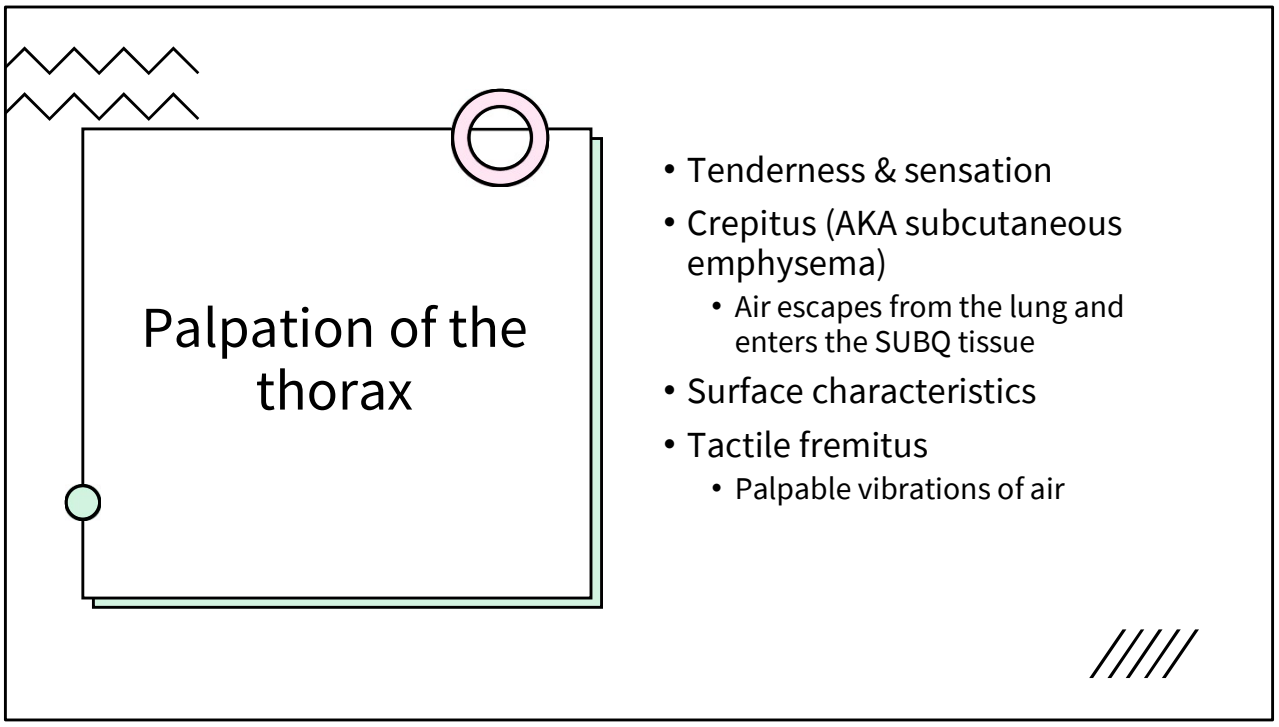
Barrel chest is caused by emphysema/COPD. Funnel chest is a congenital malformation that seldom causes symptoms other than self consciousness.

Posterior assessment of the thorax

- Inspect
  - Configuration
  - Client's positioning
- Observe
  - Use of accessory muscles



Have the client sit with arms at the sides, stand behind the client and observe the position of the scapulae and the shape and configuration: ratio 1:2.. Once again, a 1:1 ratio is abnormal: barrel chest. If you see a patient using the trapezius or shoulders muscles to facilitate inspiration, this is abnormal.
Observe use of accessory muscles: Normal: no use of accessory muscles; abnormal: client leans forward and uses arms to support weight and lift chest to increase breathing capacity….Also known as the **tripod position**. This position is used a lot in patients with COPD.
Client positioning: Normal is a client sitting up and relaxed, breathing easily with arms at sides in lap.



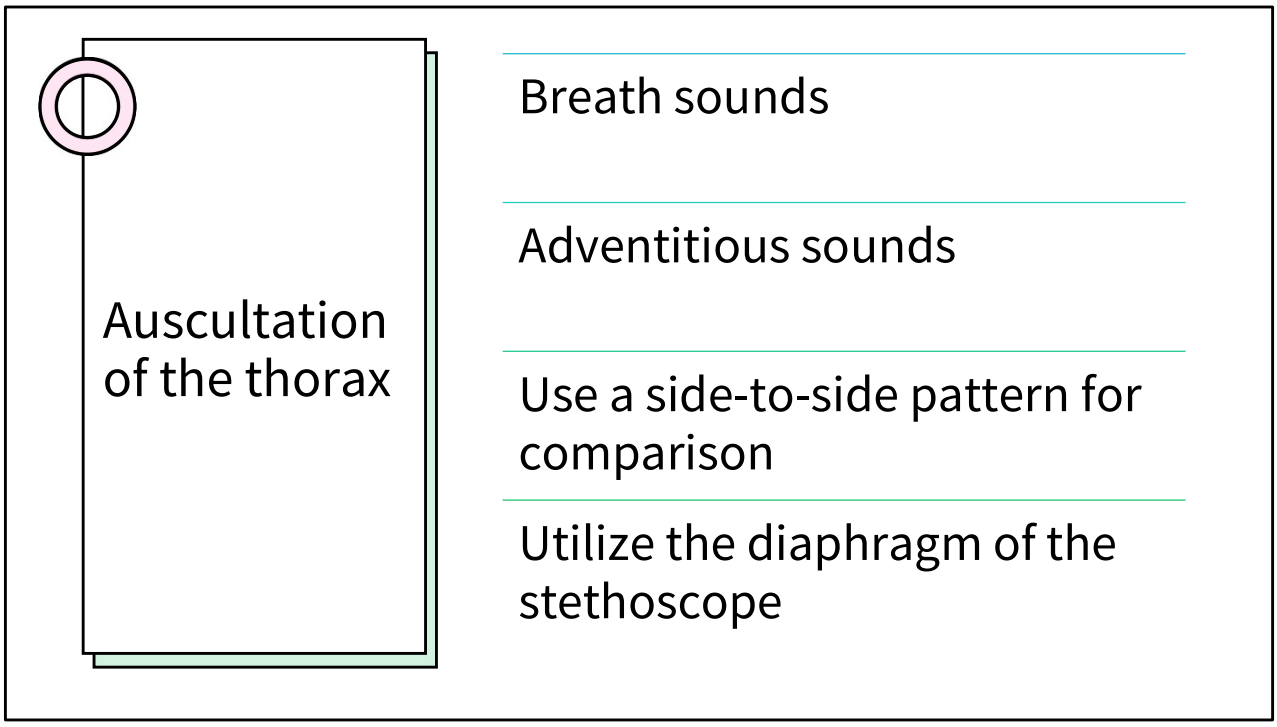
Tenderness: Normal is no tenderness, no pain, or unusual sensation. Tenderness could be from soreness from exercise or excessive work of breathing

Temperature should be equal bilaterally. Abnormal: increased warmth: may be related to local infection.

Crepitus: is subcutaneous emphysema. Air has escaped from the lungs or airways into the SUBQ tissues. Can be seen after an open thoracic injury, around a chest tube, or a tracheostomy. Feels like “rice Krispies” popping under your fingers.

Surface Characteristics: palpating for any lesions, normal findings is the skin and subcutaneous tissue are free of lesions and masses.

Tactile Fremitus: use the **ball or ulnar edge** of one hand to assess for fremitus (vibrations of air in the bronchial tubes transmitted to the chest wall). As you move your hand to each area, ask the client to say “ninety-nine”. Normal findings: fremitus is symmetric and easily identified in the upper regions of the lungs. Fremitus should remain symmetric for bilateral positions. Abnormal findings: Unequal fremitus is usually the result of consolidation (which increases fremitus) or bronchial obstruction, air trapping, in emphysema, pleural effusion, or pneumothorax.



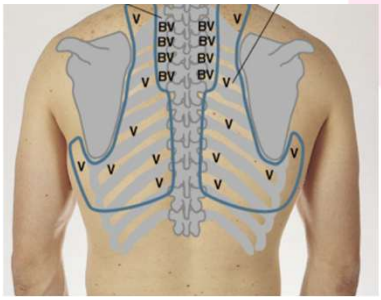
We are going to further discuss the breath sounds in other slides, but when you are listening to breath sounds, it is best to use a side-to-side pattern for comparison of each lung, does each lung sound the same in the apex? The base? And so forth.

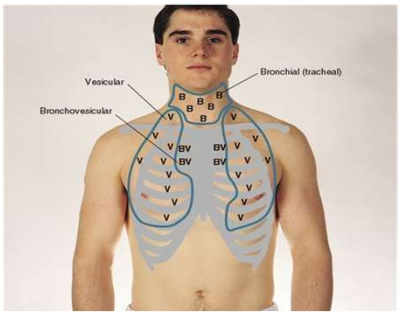
Auscultation of the thorax: Breath sounds

- Bronchial
  - High-pitched, loud, harsh, hollow
  - Short inspiration, long expiration
  - Located at the trachea and larynx
- Bronchovesicular
  - Moderate pitch, mixed quality, moderate amplitude
  - Same duration inspiration and expiration
  - Located over the major bronchi
- Vesicular
  - Low-pitched, breezy, soft
  - Long inspiration, short expiration
  - Located over the peripheral lung fields

Bronchial- tracheal area, high pitched, loud, harsh, hollow, located at trachea and larynx
Bronchovesicular – over major bronchi where fewer alveoli are located, same during inspiration and expiration, posterior around the scapulae; anterior around the upper sternum in the first and second intercostal spaces
Vesicular- the sounds are low, long in inspiration, short in expiration and heard over peripheral lung fields

LOCATION OF BREATH SOUNDS





Remember the locations of the breath sounds: Bronchial, bronchovesicular, and vesicular

https://www.youtube.com/watch?v=64bLgnv1mHA

https://www.youtube.com/watch?v=TlgP8MzlMaw&t=149s

WWW.EASYAUSCULTATION.COM

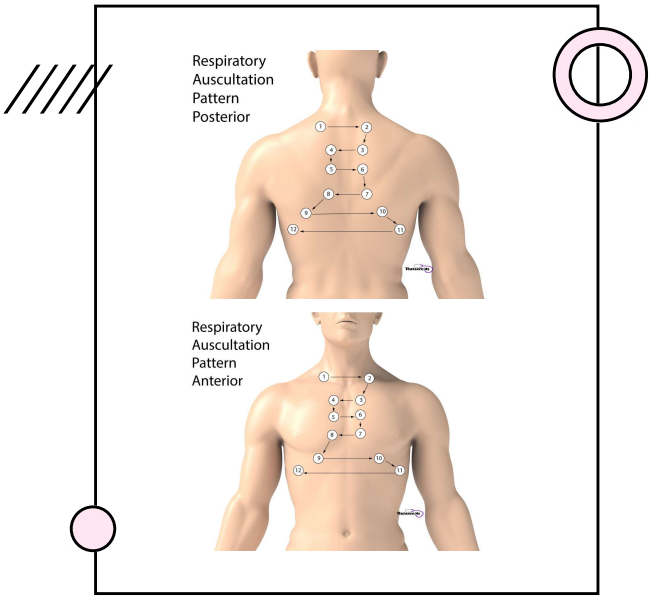
Auscultation of the thorax: Adventitious sounds

- Crackles (fine or coarse)
  - Popping sound heard during inspiration
- Wheeze (Sibilant)
  - Continuous, musical, heard during inspiration OR expiration
- Wheeze (Sonorous)
  - Rattling or snoring sound, usually in the bronchial airways due to secretions
- Pleural friction rub
  - Inflamed pleural surfaces rubbing together; sounds like leather rubbing together



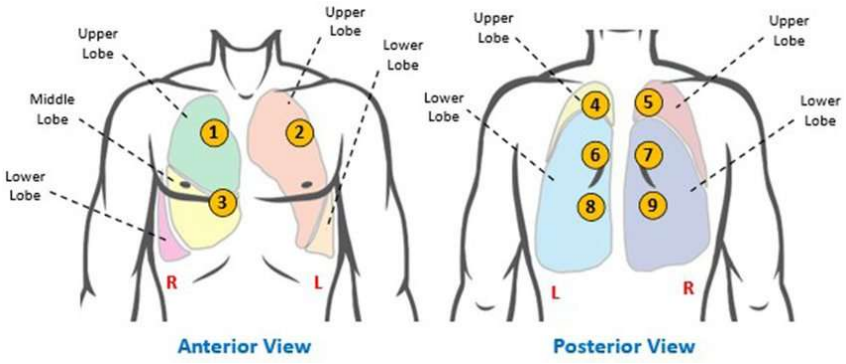
Crackles (Fine)- High-pitched popping sounds heard during inspiration. Not cleared with coughing. Air passing through moisture. Often in the lung bases. (Pneumonia, CHF, bronchitis, asthma, emphysema)
Crackles (Coarse)- Lower-pitched bubbling, moist sound. (Pneumonia, pulmonary edema, pulmonary fibrosis)
Wheeze (Sibilant)- High-pitched, continuous musical sounds, can be heard during inspiration or expiration. Air passing through constricted passages. (Asthma, emphysema)
Wheeze (Sonorous)- Low-pitched, snoring, rattling, moaning sound. Can be heard during inspiration or expiration. MAY CLEAR WITH COUGHING. (Bronchitis, or before episodes of sleep apnea)
Pleural friction rub: Low-pitched, dry, grating sound. Occurs during inspiration & expiration. From 2 pleural surfaces rubbing together. (Pleuritis)

P A T T E R N O F
A U S C U L T A T I O N



This is the side-to-side comparison model to remember when listening for lung sounds on your head-to-toe assessment

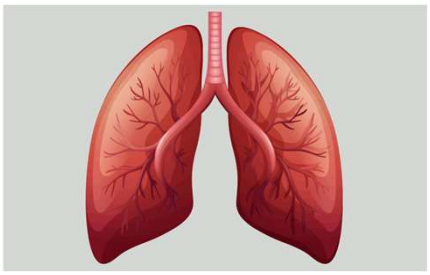
Location of lung lobes



This is a diagram shows an approximate estimate where the lobes are located on a client. The apex of the lobes are located at the clavicle and the base of the lungs is a little lower than the nipple and lateral when you are auscultating anterior and posterior the upper lungs are a smaller area compared to the base.

Auscultation of the thorax: Guidelines

- Listen in all areas for one full respiration
- Have the client breathe normally; you do not want them to hyperventilate
- ALWAYS listen over bare skin, not over clothing



Take breaks with clients that have lung disease like asthma or COPD

Health promotion & disease prevention: COPD: Chronic obstructive pulmonary disease

- Chronic inflammatory lung disease that causes obstructed airflow from the lungs
  - Airways constrict AFTER air is in the lungs, making exhaling more difficult
- 2 main conditions
  - Emphysema & chronic bronchitis
- Progressive - 3rd leading cause of death in the US
- Healthy People 2030 goal:
  - Reduce emergency department visits of COPD in adults
  - Reduce hospitalizations for COPD



3rd leading cause of death in the US
Major cause of global morbidity and mortality

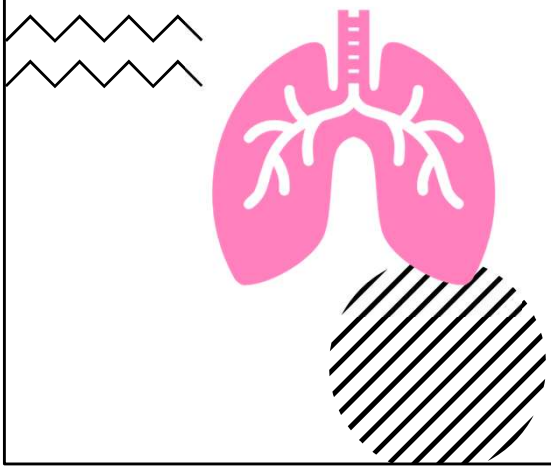
COPD

- Risks:
  - Smoking or secondhand exposure
  - Asthma
  - Occupational exposure to dust & chemicals
  - Exposure to fumes from burning fuels
- Client teaching:
  - Avoid smoking
  - Use preventative measures if exposed to occupational respiratory irritants
  - Treatments, breathing techniques, energy-conserving techniques



Pursed lip breathing reduces respirations, improves expiratory phase, and delays small airway collapse, thereby reducing air trapping in lungs.
Tripod positioning (sit or stand leaning forward with the arms supported) forces diaphragm down and forward, stabilizing chest while reducing work of breathing. If shortness of breath worsens with arms raised, support arms during ADLs.
Energy-conserving methods: Pace activities, take frequent rests, use assistive devices, and break activities into smaller tasks to help reduce dyspnea development.

COPD



- Assessment findings:
  - Barrel chest
  - Tripod positioning
  - Pursed-lip breathing
  - Diminished breath sounds
  - Adventitious breath sounds
    - Wheezes, crackles
  - Use of accessory muscles for breathing
  - Tachypnea
  - Dyspnea
  - Cyanosis
  - Clubbing

Health promotion & disease prevention: Lung cancer

- Leading cause of cancer death in the US (ACS, 2025)
- Average age of diagnosis: 70 (ACS, 2023)
- 5-year survival rate is 65% when detected when the disease is still localized within the lungs.
- More than half of people with lung cancer die within one year of being diagnosed (ALA, 2023)
- Healthy People 2030 goals:
  - Multiple focusing on decreasing lung cancer deaths and smoking cessation

2 types:
Small cell (13%) & non-small cell (84%) (ALA, 2024)

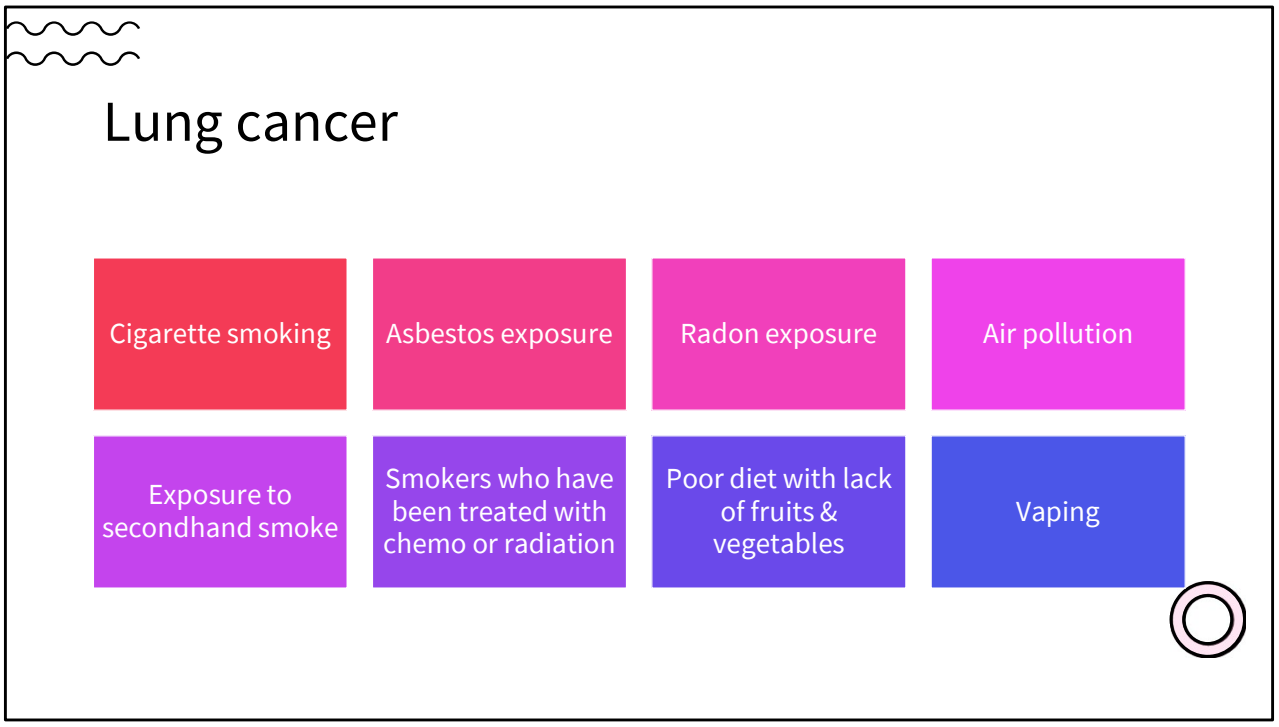
American cancer society (2023):
Lung cancer (both small cell and non-small cell) is the second most common cancer in both men and women in the United States (not counting skin cancer). In men, prostate cancer is more common, while in women breast cancer is more common.

Lung cancer mainly occurs in older people. Most people diagnosed with lung cancer are 65 or older; a very small number of people diagnosed are younger than 45. The average age of people when diagnosed is about 70.

5-year survival rate is 65% when detected when the disease is still localized within the lungs. Only 28% of lung cancers are diagnosed at this early stage. For distant tumors (spread to other organs) the 5-year survival rate is only 10% (ALA, 2026)

Lung cancer is by far the leading cause of cancer death in the US, accounting for about 1 in 5 of all cancer deaths. Each year, more people die of lung cancer than of colon, breast, and prostate cancers combined.
On a positive note, the number of new lung cancer cases continues to decrease, partly because more people are quitting smoking (or not starting). The number of deaths from lung cancer continues to drop as well, due to fewer people smoking and advances in early detection and treatment.

Lung cancer



The most primary risk factor for lung cancer is????? SMOKING 90% of lung cancer cases in the United States is from inhaling cigarette smoke and the toxic mix of over 7000 chemicals that come with cigarettes.

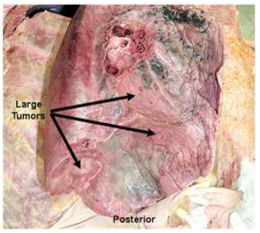
Studies have shown that many chemical groups found in vaping and the environment may disrupt pulmonary functions and initiate carcinogenesis in an additive or synergistic manner. There is strong evidence that tobacco and flavoring compounds contribute to lung cancer. Exposure to many toxins found in vaping products and the environment, such as certain metals, indicates that these substances may increase one's risk for tumorigenesis

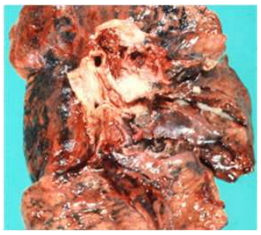


Lung cancer

- Client education:
  - Avoid smoking
  - Avoid secondhand smoke exposure
  - Have home checked for asbestos and radon
  - Wear protective gear for occupational exposure to hazards
  - Seek evaluation for symptoms such as prolonged cough or chest pain

WHAT LUNG CANCER LOOKS LIKE







Thoracotomy… client had either their whole lung or part of their lung removed.

Review question

- The purpose of the side-to-side comparison used during respiratory assessment is to:

- A. Prevent fatiguing the client
- B. Tightly organize the assessment
- C. Allow for comparison of like areas
- D. Conserve time



C

Review question

- Crackles are:

- A. Caused by air passing through moisture
- B. Heard most often during inspiration
- C. Found in lung bases with congestive heart failure
- D. All of the above



D